

PROJECT REPORT: DATA-DRIVEN MARKET EXPANSION & LEAD PRIORITIZATION

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Target Audience: VP of Sales, Head of Business Development, Chief Commercial Officer

1. EXECUTIVE SUMMARY

In highly competitive B2B landscapes, static outbound prospecting creates high operational overhead, elongated sales cycles, and bloated Customer Acquisition Costs (CAC). This project builds an automated analytics engine using historical transaction records from 'sales_data_sample.csv' to transform the Business Development (BD) pipeline. By pairing unsupervised machine learning (K-Means Clustering) with supervised predictive models (Random Forest Classifier), we successfully segmented the market, isolated our corporate Ideal Customer Profile (ICP), managed real-world class imbalances, and established a dynamic resource allocation strategy that maximizes Net Revenue Retention (NRR).

2. PROBLEM STATEMENT & BUSINESS CONTEXT

The organization seeks strategic expansion into alternative market channels and territories with a finite number of BD personnel. The legacy sales pipeline suffers from untargeted outbound effort where sales representatives spend identical hours chasing low-yield accounts versus enterprise-tier targets, leading to pipeline information asymmetry and revenue leakage.

3. DATA CLEANING & METRIC FEATURE ENGINEERING

Exploratory Data Analysis (EDA) on the primary transaction set revealed incomplete geographic indicators. To identify customer values rather than order values, individual item logs were aggregated by unique CUSTOMERNAME keys to engineer three distinct commercial KPIs: Total Revenue, Order Frequency, and Total Volume.

4. MARKET SEGMENTATION (UNSUPERVISED ALGORITHMIC MODELING)

Because numerical variables vary radically in scale, features were normalized using standard scaling. A K-Means Clustering algorithm was applied to define target tiers.

Cluster 2 (Tier 1 - The Corporate ICP): This segment represents high-spending, high-frequency enterprise clients (e.g., AV Stores, Co., Anna's Decorations, Ltd, Australian Collectors, Co.). These clients order consistently and generate peak baseline revenues.

Cluster 0 & 1 (Tier 2 & 3 - Market Run-Rate): These profiles display low-to-medium transactional velocity and lower total volume.

5. LEAD SCORING & MACHINE LEARNING ITERATIONS

To predict the categorical classification of incoming deals, a machine learning framework was constructed.

Iteration 1 (Baseline): The raw target variable displayed extreme class imbalance. The baseline model returned a misleading overall accuracy (~92%), but suffered an abysmal Recall of 17% and Precision of 33% for Class 1 (Large Deals).

Iteration 2 (Optimization): To resolve the class imbalance, the classifier was updated with class_weight='balanced' and the operational decision boundary was shifted down from 50% probability to 25% probability. Shifting the activation threshold to 25% boosted the Recall for large deals to 39%—more than doubling the volume of enterprise deals flagged by the pipeline.

6. STRATEGIC RECOMMENDATIONS & BUSINESS DEVELOPMENT EXECUTION

A. The 80/20 Pareto Resource Allocation: Direct core account executives and outbound sales development tools exclusively toward profiles mirroring Cluster 2 parameters. Retain a lean run-rate team to handle inbound volumes and nurture accounts residing within Clusters 0 and 1.

B. Dynamic Quarterly Recalibration Checkpoints: The analytics system will run automated checks on a rolling 90-day cycle. If order frequencies within a Cluster 0 or 1 client profile surpass predefined statistical threshold triggers, the system will automatically upgrade that account to Tier 1 status.

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Project Title: Data-Driven Market Expansion & Lead Prioritization System

Document Version: 1.0

Date: May 2026

1. PROJECT OVERVIEW & OBJECTIVES

The objective of this initiative is to build an automated lead categorization and scoring pipeline that ingests commercial transaction attributes, clusters corporate customer value tiers, and generates algorithmic scores to prioritize active sales pipelines.

2. STAKEHOLDER CONTEXT & RESPONSIBILITIES

Business Development / Sales Managers: End-users who utilize the output rankings to target active prospects.

Data Science / Analytics Team: Product owners responsible for algorithm deployment and model training.

Sales Operations / BI Engineers: Engineers responsible for ingesting pipeline inputs into CRM frameworks.

3. FUNCTIONAL REQUIREMENTS MATRIX

REQ ID	Category	Description	Priority (MoSCoW)
FR-01	Data Ingestion	Ingest multi-variable customer logs.	Must Have
FR-02	Customer Clustering	Run unsupervised K-Means model to tier accounts.	Must Have
FR-03	Weighted Classification	Utilize balanced class-weights for rare deals.	Must Have
FR-04	Probability Score	Calculate continuous confidence thresholds (0-100%).	Must Have
FR-05	Threshold Trigger	Trigger alerts whenever enterprise probability score >= 25%.	Should Have

FR-06	BI Sales Dashboard	Render dynamic visual layout highlighting 'Hot Leads'.	Should Have
FR-07	Recalibration	Automatically re-cluster data groups on a 90-day cycle.	Could Have

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Scalability: Process batch uploads up to 10,000 corporate line items in < 60 seconds.

4.2 Governance: Missing address values must not crash the calculation pipeline; fallback logic must aggregate by country codes.

5. BUSINESS DEPLOYMENT

The sales floor will structure outbound work according to a 25% threshold rule: leads with an enterprise probability $\geq 25\%$ are classified as Tier 1 Hot Enterprise Leads and receive 80% of BD allocation. Accounts below this threshold receive 20% of allocation and are monitored for quarterly recalibration based on purchase frequency triggers.